

GoldenHour

Database Connection Documentation

MySQL 8 — Schema, Connection, and Configuration

1. Database Connection

The backend connects to MySQL 8 using the mysql2/promise library with a connection pool.

```
DB_HOST=localhost DB_USER=root DB_PASSWORD=your_mysql_password DB_NAME=goldenhour
DB_PORT=3306
```

Connection Pool Configuration:

```
const pool = mysql.createPool({ host: process.env.DB_HOST, user: process.env.DB_USER,
password: process.env.DB_PASSWORD, database: process.env.DB_NAME, port:
process.env.DB_PORT || 3306, waitForConnections: true, connectionLimit: 10, queueLimit: 0
});
```

2. Database Schema Overview

Database: goldenhour | Engine: InnoDB | Charset: utf8mb4 | Tables: 13

Table	Purpose	Key Columns
hospitals	Registered hospitals	hospital_id, name, latitude, longitude, contact
users	All system users	user_id, email, role, hospital_id, ambulance_id
ambulances	Registered ambulances	ambulance_id, registration_number, status
cases	Core case lifecycle	case_id, case_code, status, priority, eta_minutes
case_clinical_data	ATMIST handover data	case_id, gcs, spo2, bp, pulse, injuries
critical_flags	Emergency flags	case_id, shock, hypoxia, cardiac_arrest
location_updates	Live GPS history	case_id, latitude, longitude, eta_minutes
acknowledgements	Hospital acceptance	case_id, acknowledged_by, acknowledged_at
team_readiness	Trauma team prep	case_id, trauma_team_activated, roles_assigned
hospital_capacity	ED capacity status	hospital_id, resus_bays, ct_available, diversion
resource_requests	Resource needs	case_id, blood_required, imaging_required
activity_events	Immutable audit log	case_id, event_type, event_data, performed_at
notifications	Alerts sent	case_id, recipient_user_id, channel, sent_at

3. Core Table — cases

Column	Type	Description
case_id	INT PK AUTO_INCREMENT	Unique case identifier
case_code	VARCHAR(20) UNIQUE	Human-readable code e.g. GH-2026-0001
ambulance_id	INT FK	References ambulances table
destination_hospital_id	INT FK	References hospitals table
priority	ENUM	CRITICAL / HIGH / MEDIUM / LOW
status	ENUM	DRAFT→SENT→DELIVERED→ACKNOWLEDGED→ARRIVED→CLOSED

age_band	VARCHAR(20)	Patient age band e.g. 40-49
sex	ENUM	M / F / OTHER
chief_complaint	TEXT	Primary complaint/case type
eta_minutes	INT	Estimated time of arrival in minutes
latitude	DECIMAL(10,8)	Ambulance latitude
longitude	DECIMAL(11,8)	Ambulance longitude
created_by	INT FK	References users table
created_at	TIMESTAMP	Auto-set on creation
updated_at	TIMESTAMP	Auto-updated on change

4. hospitals Table

Column	Type	Description
hospital_id	INT PK AUTO_INCREMENT	Unique hospital ID
name	VARCHAR(150)	Hospital name
address	TEXT	Full address
latitude	DECIMAL(10,8)	GPS latitude for Haversine matching
longitude	DECIMAL(11,8)	GPS longitude for Haversine matching
contact	VARCHAR(20)	Contact number shown to ambulance crew
created_at	TIMESTAMP	Auto-set on creation

5. Seed Data

The following hospitals are pre-loaded in the database:

ID	Name	Location	Contact
1	City Emergency Hospital	MG Road, Bangalore	080-12345678
2	Apollo Hospital	Bannerghatta Road, Bangalore	080-87654321
3	Manipal Hospital	Old Airport Road, Bangalore	+918025023700
4	Victoria Hospital	Bowring Road, Bangalore	080-22206439

6. Entity Relationships

users → cases (one user creates many cases)

ambulances → cases (one ambulance has many cases)

hospitals → cases (one hospital receives many cases)

cases → case_clinical_data (one case has one clinical record)

cases → critical_flags (one case has one flags record)

cases → location_updates (one case has many location updates)

cases → acknowledgements (one case has one acknowledgement)

cases → team_readiness (one case has one readiness record)

cases → activity_events (one case has many audit events)

hospitals → hospital_capacity (one hospital has one capacity record)

7. Case Status Flow

DRAFT → SENT → DELIVERED → ACKNOWLEDGED → ARRIVED → CLOSED

Exception states:

Status	Meaning
QUEUED	Saved offline, waiting for network
DIVERTED	Redirected to different hospital
ACK_FAILURE	No acknowledgement within timeout
CANCELLED	Case cancelled by crew